CO2-AT-3

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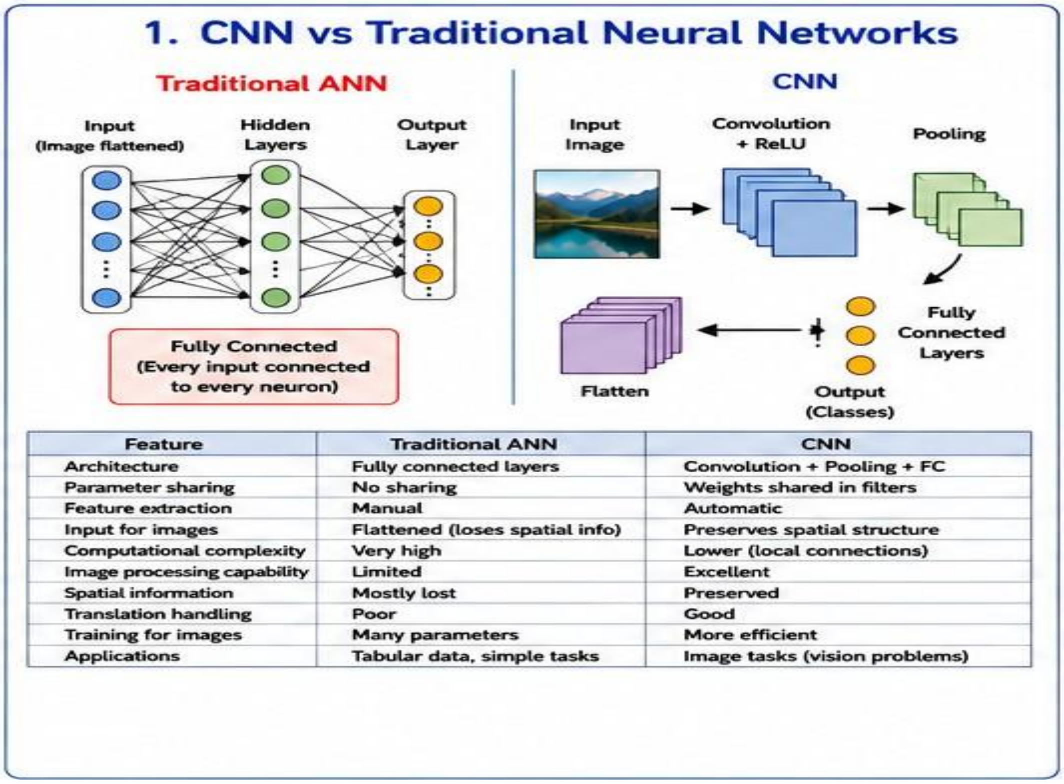
COURSE CODE:MLA0401
COURSE NAME:DEEP LEARNING

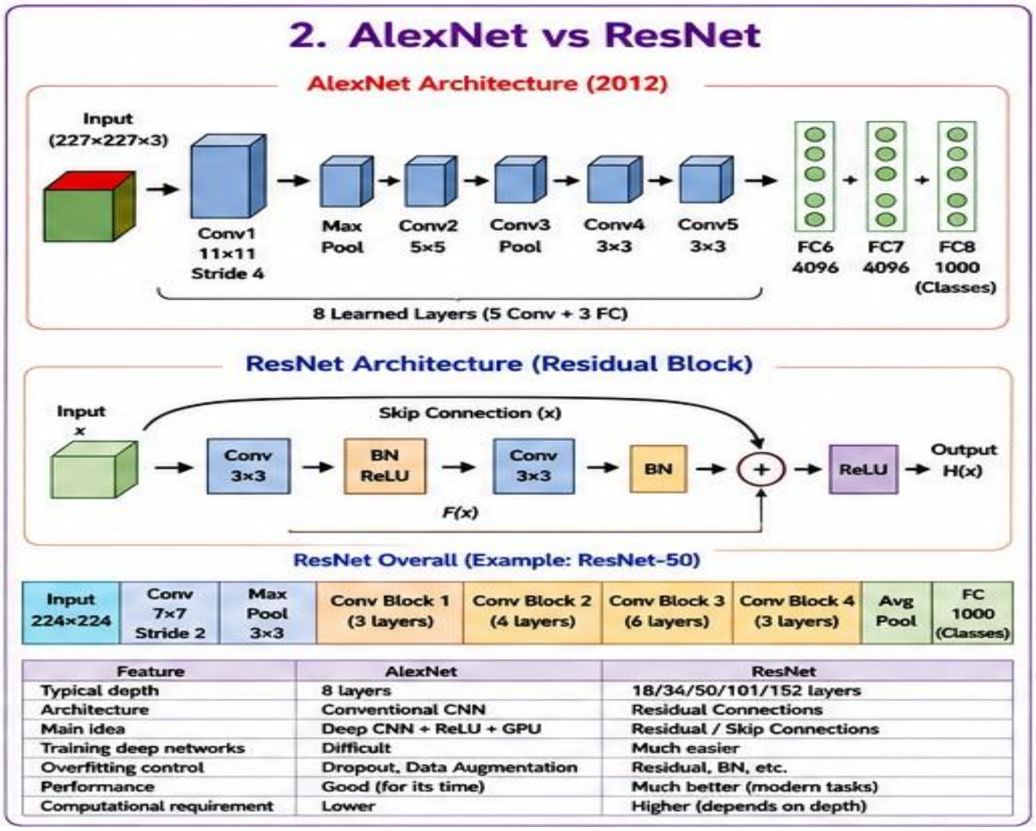
1. CNN vs Traditional ANN



Point	**ANN**	**CNN**
Architecture	Fully connected	Convolution + Pooling
Parameters	More	Fewer
Feature extraction	Manual	Automatic
Image processing	Less suitable	Highly suitable
Spatial information	Lost	Preserved
Parameter sharing	No	Yes
Speed for images	Slower	Faster

Applications General data Images/videos

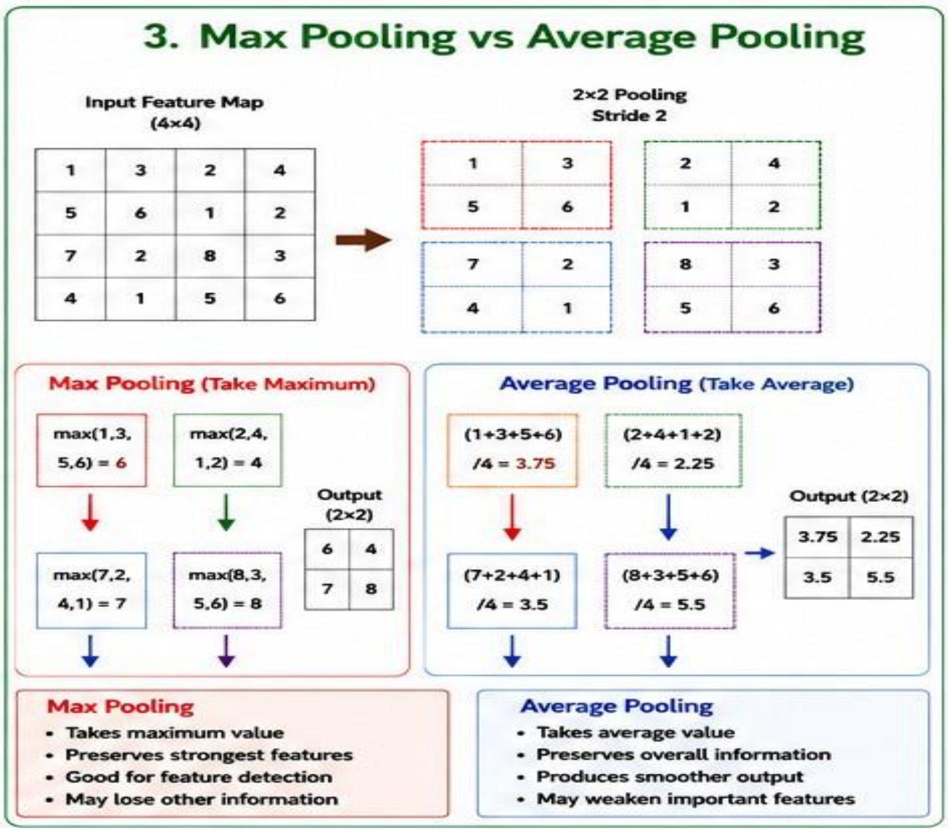
2. AlexNet vs ResNet



Point	**AlexNet**	**ResNet**
Year	2012	2015
Layers	8	18–152+
Architecture	Normal CNN	Residual CNN
Skip connection	No	Yes
Training	Difficult for deep networks	Easier

Main feature	ReLU + GPU	Residual learning
Performance	Good	Better
Computation	Lower	Higher for deeper models

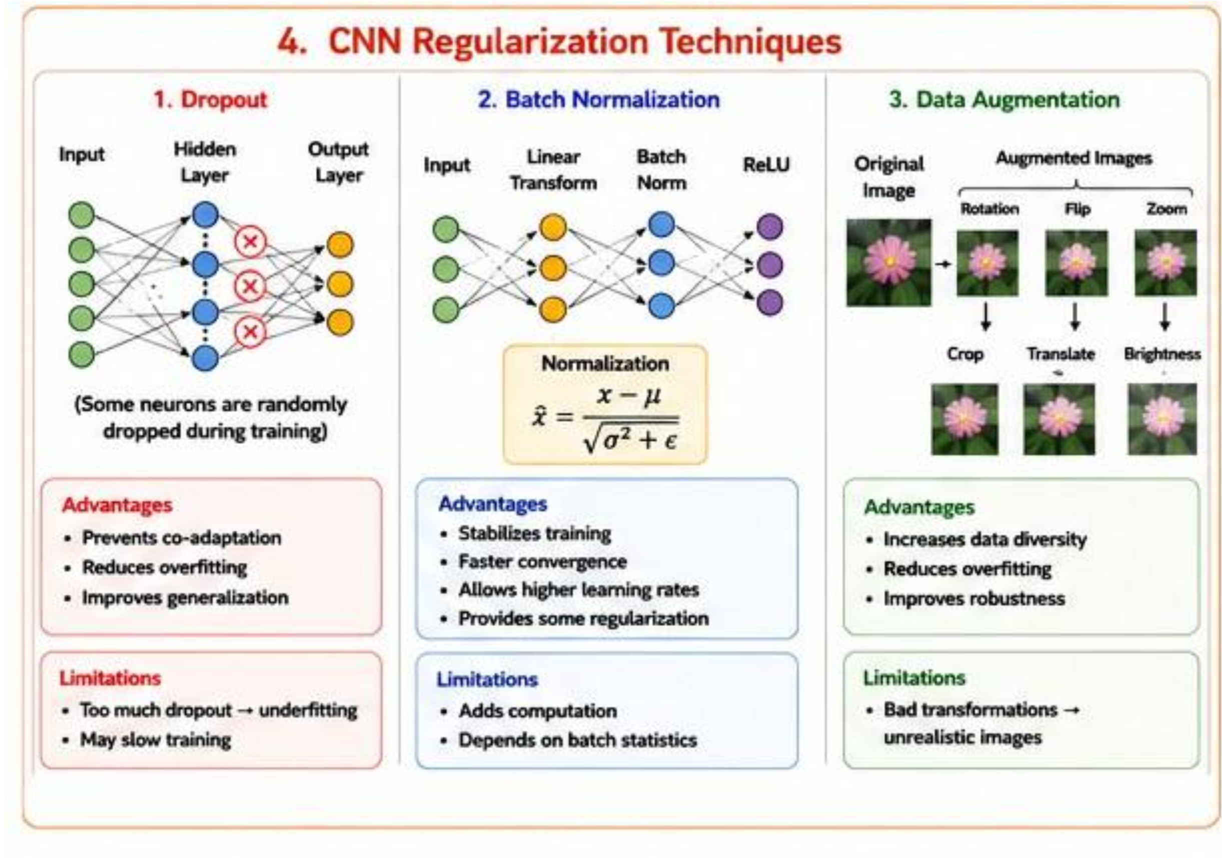
3. Max Pooling vs Average Pooling



Point	**Max Pooling**	**Average Pooling**
Operation	Maximum value	Average value
Feature	Strongest feature	Overall feature
Output	Sharper	Smoother
Advantage	Keeps important features	Keeps overall information

Disadvantage	Loses some information	May weaken features
Common use	Image classification	Feature smoothing
Example	max(1,3,5,6) = **6**	(1+3+5+6)/4 = **3.75**

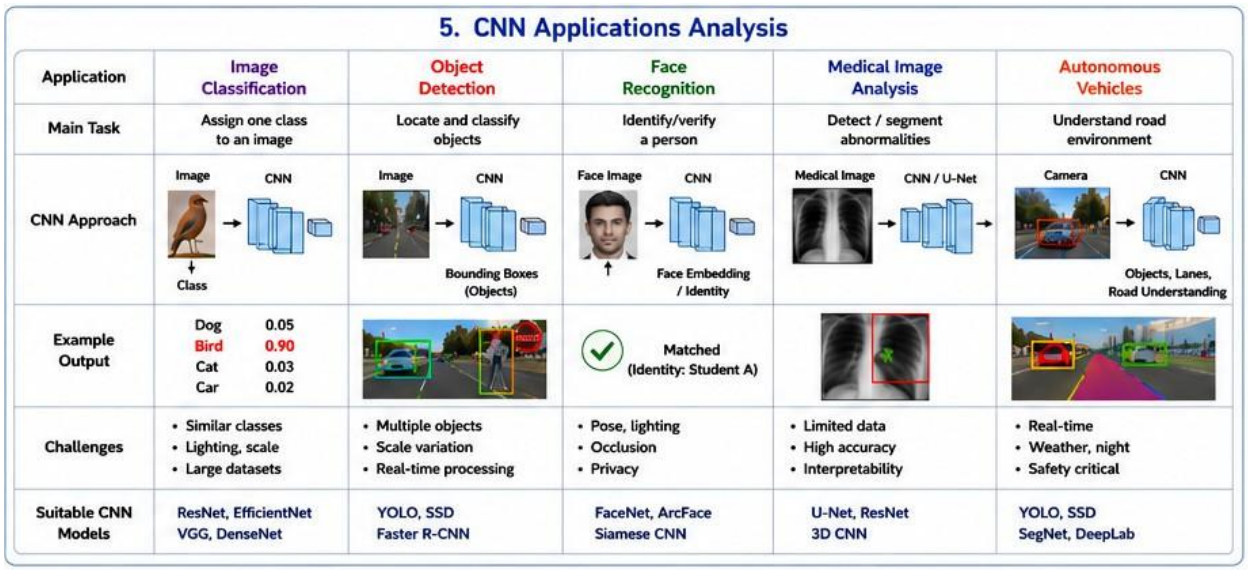
4. Dropout vs Batch Normalization vs Data Augmentation



Point	Dropout	Batch Normalization	Data Augmentation
Purpose	Reduce overfitting	Stable training	Increase data
Method	Removes neurons randomly	Normalizes activations	Modifies images
Example	Dropout = 0.5	Mean/variance normalization	Rotate, crop, zoom

Main benefit	Better generalization	Faster training	More diverse data
Limitation	Too much → underfitting	Needs batch statistics	Wrong changes can hurt
Applied to	Network	Network layers	Input images

5. CNN Applications



5. CNN Applications Analysis

Application	Image Classification	Object Detection	Face Recognition	Medical Image Analysis	Autonomous Vehicles
Main Task	Assign one class to an image	Locate and classify objects	Identify/verify a person	Detect / segment abnormalities	Understand road environment
CNN Approach	Image → CNN → Class	Image → CNN → Bounding Boxes (Objects)	Face Image → CNN → Face Embedding / Identity	Medical Image → CNN / U-Net	Camera → CNN → Objects, Lanes, Road Understanding
Example Output	Dog 0.05, Bird 0.90, Cat 0.03, Car 0.02		Matched (Identity: Student A)		
Challenges	• Similar classes • Lighting, scale • Large datasets	• Multiple objects • Scale variation • Real-time processing	• Pose, lighting • Occlusion • Privacy	• Limited data • High accuracy • Interpretability	• Real-time • Weather, night • Safety critical
Suitable CNN Models	ResNet, EfficientNet VGG, DenseNet	YOLO, SSD Faster R-CNN	FaceNet, ArcFace Siamese CNN	U-Net, ResNet 3D CNN	YOLO, SSD SegNet, DeepLab

Application	**Purpose**	**Example CNN**
Image Classification	Classify image	ResNet
Object Detection	Find + locate objects	YOLO
Face Recognition	Identify person	Face CNN
Medical Analysis	Detect diseases	U-Net, ResNet
Autonomous Vehicles	Detect road objects	YOLO, SSD